SCHOOL OF COMPUTER SCIENCE AND IT
YBN UNIVERSITY RANCHI

JAVA PROGRAMMING
ASSIGNMENT 02

1) Check if Two Integers are Equal

Description: Write a Java program to accept two integers and check whether they are equal.

Test Data: `15 15`

Expected Output: `Number1 and Number2 are equal`

Hint:

- Use the `==` operator to compare the two integers.
- If the integers are equal, display a message indicating they are equal; otherwise, indicate they are not equal.
- Use the `Scanner` class for input.

2) Check if a Number is Even or Odd

Description: Write a Java program to check whether a given number is even or odd.

Test Data: `15`

Expected Output: `15 is an odd integer`

Hint:

- Use the modulus operator `%` to check if the number is divisible by 2.
- If `number % 2 == 0`, the number is even; otherwise, it is odd.
- Use the `Scanner` class for input.

3) Check if a Number is Positive or Negative

Description: Write a Java program to check whether a given number is positive or negative.

Test Data: `15`

Expected Output: `15 is a positive number`

Hint:

- Use an `if-else` statement to check if the number is greater than, less than, or equal to 0.
- Display the appropriate message based on the comparison.

- Use the `Scanner` class for input.

4) Check if a Year is a Leap Year

Description: Write a Java program to find whether a given year is a leap year or not. **Test**

Data: `2016`

Expected Output: `2016 is a leap year.`

Hint:

- A year is a leap year if it is divisible by 4, but not by 100 unless it is also divisible by 400.
- Use nested `if` statements to implement this logic.
- Use the `Scanner` class for input.

5) Check Voting Eligibility

Description: Write a Java program to read the age of a candidate and determine whether they are eligible to vote.

Test Data: `21`

Expected Output: `Congratulation! You are eligible for casting your vote.`

Hint:

- Use an `if` statement to check if the age is greater than or equal to 18.
- Display a congratulatory message if the candidate is eligible, otherwise indicate they are not.
- Use the `Scanner` class for input.

6) Display Value of n Based on m

Description: Write a Java program to read the value of an integer `m` and display the value of `n` as follows:

- `n = 1` if `m > 0`
- `n = 0` if `m == 0`
- `n = -1` if `m < 0`

  Test Data: `-5`

  Expected Output: `The value of n = -1`

  Hint:

- Use `if-else if-else` statements to compare the value of `m` and assign the appropriate value to `n`.
- Use the `Scanner` class for input.

7) Categorize a Person Based on Height

Description: Write a Java program to accept the height of a person in centimeters and categorize them as follows:

- Dwarf if height < 150 cm
- Average height if height is between 150 cm and 165 cm
- Tall if height is greater than 165 cm

Test Data: `135`

Expected Output: `The person is Dwarf.`

Hint:

- Use `if-else if-else` statements to compare the height and display the appropriate category.
- Use the `Scanner` class for input.

8) Find the Largest of Three Numbers

Description: Write a Java program to find the largest of three numbers.

Test Data: `12 25 52`

Expected Output:

```
1st Number = 12,
2nd Number = 25,
3rd Number = 52

The 3rd Number is the greatest among three
```

Hint:

- Use nested `if-else` statements or the ternary operator `?:` to compare the three numbers.
- Display the largest number.
- Use the `Scanner` class for input.

9) Determine the Quadrant of a Coordinate Point

Description: Write a Java program to accept a coordinate point in an XY coordinate system and determine in which quadrant the point lies.

Test Data: `7 9`

Expected Output: `The coordinate point (7,9) lies in the First quadrant.`

Hint:

- Use `if-else if-else` statements to check the signs of the `x` and `y` coordinates and determine the quadrant.
- Use the `Scanner` class for input.

10) Check Eligibility for Admission

Description: Write a Java program to determine eligibility for admission to a professional course based on the following criteria:

- Marks in Maths ≥ 65, Marks in Physics ≥ 55, and Marks in Chemistry ≥ 50
- Total marks in all three subjects ≥ 190 or total marks in Maths and Physics ≥ 140

Test Data:

```
Input the marks obtained in Physics : 65
Input the marks obtained in Chemistry : 51
Input the marks obtained in Mathematics : 72
Total marks of Maths, Physics, and Chemistry : 188
Total marks of Maths and Physics : 137
```

Expected Output: `The candidate is not eligible for admission.`

Hint:

- Use nested `if-else` statements to check all conditions for eligibility.
- Display whether the candidate is eligible or not based on the criteria.
- Use the `Scanner` class for input.